

Rajshahi University of Engineering & Technology

Department of Computer Science & Engineering

CSE 3102, Lab No. 10, Lab Work

Follow the instructions below and provide a report based on the results of each step. Begin by completing the initial tasks to set up the database and create the necessary tables.

Database Name: MovieBd

Step 1: Create the following tables with appropriate constraints:

1. Movies Table

```
CREATE TABLE Movies (  
    movie_id INT PRIMARY KEY AUTO_INCREMENT,  
    movie_title VARCHAR(100) NOT NULL,  
    genre VARCHAR(50) NOT NULL,  
    release_year INT CHECK (release_year >= 1888), -- first film was in 1888  
    rating DECIMAL(3,1) CHECK (rating >= 0 AND rating <= 10),  
    runtime_minutes INT,  
    director_id INT,  
    FOREIGN KEY (director_id) REFERENCES Directors(director_id)  
);
```

2. Actors Table

```
CREATE TABLE Actors (  
    actor_id INT PRIMARY KEY AUTO_INCREMENT,  
    actor_name VARCHAR(100) NOT NULL,  
    birth_date DATE,  
    gender VARCHAR(10) CHECK (gender IN ('Male', 'Female', 'Other'))
```

);

3. Movie_Actors Table (Many-to-Many)

```
CREATE TABLE Movie_Actors (  
    movie_id INT,  
    actor_id INT,  
    role_name VARCHAR(100),  
    PRIMARY KEY (movie_id, actor_id),  
    FOREIGN KEY (movie_id) REFERENCES Movies(movie_id),  
    FOREIGN KEY (actor_id) REFERENCES Actors(actor_id)  
);
```

4. Directors Table

```
CREATE TABLE Directors (  
    director_id INT PRIMARY KEY AUTO_INCREMENT,  
    director_name VARCHAR(100) NOT NULL,  
    birth_date DATE  
);
```

5. Awards Table

```
CREATE TABLE Awards (  
    award_id INT PRIMARY KEY AUTO_INCREMENT,  
    award_name VARCHAR(100) NOT NULL,  
    year INT NOT NULL,  
    movie_id INT,  
    actor_id INT,  
    category VARCHAR(100),  
    FOREIGN KEY (movie_id) REFERENCES Movies(movie_id),
```

```
FOREIGN KEY (actor_id) REFERENCES Actors(actor_id)
);
```

Step 2: Insert the following sample data

1. Insert into Directors

```
INSERT INTO Directors (director_name, birth_date) VALUES
('Christopher Nolan', '1970-07-30'),
('Quentin Tarantino', '1963-03-27'),
('James Cameron', '1954-08-16');
```

2. Insert into Movies

```
INSERT INTO Movies (movie_title, genre, release_year, rating,
runtime_minutes, director_id) VALUES
('Inception', 'Science Fiction', 2010, 8.8, 148, 1),
('Pulp Fiction', 'Crime', 1994, 8.9, 154, 2),
('Titanic', 'Romance', 1997, 7.8, 195, 3);
```

3. Insert into Actors

```
INSERT INTO Actors (actor_name, birth_date, gender) VALUES
('Leonardo DiCaprio', '1974-11-11', 'Male'),
('Kate Winslet', '1975-10-05', 'Female'),
('John Travolta', '1954-02-18', 'Male'),
('Samuel L. Jackson', '1948-12-21', 'Male');
```

4. Insert into Movie_Actors

```
INSERT INTO Movie_Actors (movie_id, actor_id, role_name) VALUES
(1, 1, 'Cobb'),      -- Inception
(2, 3, 'Vincent Vega'), -- Pulp Fiction
```

```
(2, 4, 'Jules Winnfield'),  
(3, 1, 'Jack Dawson'), -- Titanic  
(3, 2, 'Rose DeWitt Bukater');
```

5. Insert into Awards

```
INSERT INTO Awards (award_name, year, movie_id, actor_id, category)  
VALUES  
  
('Best Picture', 1998, 3, NULL, 'Movie'),  
('Best Actor', 2011, 1, 1, 'Actor'),  
('Best Supporting Actor', 1995, 2, 4, 'Actor');
```

Step 3: Write queries to solve the following:

1. Display all movies.
2. List all actors born after 1980.
3. Find all roles with screen time more than 100 minutes.
4. Count the number of movies in each genre.
5. Show each actor with total screen time.
6. Find actors who acted in more than one movie.
7. Find movies where Leonardo DiCaprio acted.
8. Show average screen time per movie.
9. List top 3 actors with highest total screen time.
10. List genres along with number of unique actors participated.

Solution :

1. Display all movies

sql

```
SELECT * FROM Movies;
```

2. List all actors born after 1980

sql

```
SELECT * FROM Actors  
WHERE birth_date > '1980-01-01';
```

3. Find all roles with screen time more than 100 minutes

sql

```
SELECT actor_id, movie_id, role_name, screen_time  
FROM Movie_Actors  
WHERE screen_time > 100;
```

4. Count the number of movies in each genre

sql

```
SELECT genre, COUNT(*) AS movie_count  
FROM Movies  
GROUP BY genre;
```

5. Show each actor with total screen time

sql

```
SELECT A.actor_name, SUM(MA.screen_time) AS total_screen_time
FROM Actors A
JOIN Movie_Actors MA ON A.actor_id = MA.actor_id
GROUP BY A.actor_name;
```

6. Find actors who acted in more than one movie

sql

```
SELECT A.actor_name, COUNT(DISTINCT MA.movie_id) AS movie_count
FROM Actors A
JOIN Movie_Actors MA ON A.actor_id = MA.actor_id
GROUP BY A.actor_name
HAVING movie_count > 1;
```

7. Find movies where Leonardo DiCaprio acted

sql

```
SELECT M.movie_title
FROM Movies M
JOIN Movie_Actors MA ON M.movie_id = MA.movie_id
JOIN Actors A ON A.actor_id = MA.actor_id
WHERE A.actor_name = 'Leonardo DiCaprio';
```

8. Show average screen time per movie

sql

```
SELECT M.movie_title, AVG(MA.screen_time) AS avg_screen_time
FROM Movies M
JOIN Movie_Actors MA ON M.movie_id = MA.movie_id
GROUP BY M.movie_title;
```

9. List top 3 actors with highest total screen time

sql

```
SELECT A.actor_name, SUM(MA.screen_time) AS total_screen_time
FROM Actors A
JOIN Movie_Actors MA ON A.actor_id = MA.actor_id
GROUP BY A.actor_name
ORDER BY total_screen_time DESC
LIMIT 3;
```

10. List genres along with number of unique actors participated

sql

```
SELECT M.genre, COUNT(DISTINCT MA.actor_id) AS unique_actors
FROM Movies M
JOIN Movie_Actors MA ON M.movie_id = MA.movie_id
GROUP BY M.genre;
```